

Fernando Lucatelli Nunes

Researcher in Mathematics and Computer Science

Appointed Assistant Professor, University of Coimbra (to begin in 2026)



arXiv preprints: https://arxiv.org/a/lucatellinunes_f_1.html

Google Scholar

Scopus ID 57188838707

Mathematics Genealogy

Webpage: <https://fmlucatelli.github.io/>

E-mail: fernandolucatellinunes@gmail.com

PhD IN MATHEMATICS (AWARDED 2018)

10/2012 – 01/2018

University of Coimbra, PhD in Mathematics

- Thesis: *Pseudomonads and Descent* (awarded *summa cum laude*).
- Supervisor: Prof. Maria Manuel Clementino.
- Examination Committee: Prof. Martin Hyland, Dr. Marco Mackaay, Prof. Lurdes Sousa, and Prof. Peter Beier Gothen.
- Main contributions: Biadjoint triangle theorems; freely generated categorical structures; theories of presentation and coherence for low- and higher-dimensional categorical structures; Grothendieck descent theory; categorical Galois theories; low- and higher-dimensional weak limits.

PhD IN COMPUTER SCIENCE (Expected April 2026)

11/2020 – 11/2025

Utrecht University, PhD in Computer Science

- Thesis: *Freely Generated Categorical Structures and Automatic Differentiation*.
- Supervisors: Prof. Gabriele Keller and Dr. Matthijs Vákár.
- Examination Committee: Prof. Ieke Moerdijk, Prof. Bart Jacobs, Prof. Sam Staton, Prof. Robin Cockett, and Prof. Patricia Johann.
- Main contributions: Denotational semantics of programming languages; automatic differentiation; principled program transformations; recursion and iteration; categorical semantics; logical relations; type systems and dependent types; Grothendieck constructions; extensive categories.

Academic Positions

ACADEMIC APPOINTMENTS

2026 – present	Research Fellow , School of Computer Science, University of Birmingham, UK.
2025	Postdoctoral Fellow , Department of Information and Computing Sciences, Utrecht University, The Netherlands.
2023	Postdoctoral Fellow , York University, Canada.
2023	Fields Research Fellow , Fields Institute for Research in Mathematical Sciences, Canada.
2022	Oberwolfach Leibniz Fellow , Mathematisches Forschungsinstitut Oberwolfach, Germany. (Independent research fellowship; invited and coordinated a focused research programme with international collaborators.)
2018 – 2021	Research Fellow , University of Coimbra, Portugal.
2018 – 2019	Invited Assistant Professor , University of Coimbra, Portugal.
2018	Postdoctoral Fellow , Centre for Mathematics, University of Coimbra, Portugal.

VISITING ACADEMIC APPOINTMENTS

2018, 2023, 2024, 2025	Visiting Researcher , UCLouvain, Belgium.
2025	Visiting Researcher , IMPA, Rio de Janeiro, Brazil.
2019, 2025	Visiting Researcher , University of Cape Town, South Africa.
2019, 2025	Visiting Researcher , Stellenbosch University, South Africa.

RESEARCH GROUPS AND AFFILIATIONS

2025 – present	Theory of Computation , School of Computer Science, University of Birmingham, UK.
2020 – 2026	Software Technology Group , Utrecht University, The Netherlands.
2020 – 2024	External Researcher , Centre for Mathematics (CMUC), University of Coimbra, Portugal.
2013 – present	Coimbra Category Theory Group , University of Coimbra, Portugal.

DOCTORAL APPOINTMENTS

2020 – 2025	Promovendus , Utrecht University, The Netherlands.
2012 – 2018	Doctoral Researcher , University of Coimbra, Portugal.

EARLY ACADEMIC EXPERIENCE

2006 – 2008	Secondary Mathematics Teacher , Curso Dínatos, Brasília, Brazil.
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Research

RESEARCH INTERESTS

Expertise and research interests: Mathematics and Computer Science, centred on the semantic and structural foundations of computation: (1) low- and higher-dimensional universal algebra and generalised (multi)categorical structures; (2) categorical methods in logic, topology, and algebra; (3) programming language semantics and formal methods, including rewriting systems, program analysis, and principled program transformations.

SELECTED PUBLICATIONS

- Lucatelli Nunes, F., Vákár, M.: **Monoidal closure of Grothendieck constructions via Σ -tractable monoidal structures and Dialectica formulas.** Theory Appl. Categ. 44, 1153–1217. 2025. (*65 pages*).
- Lucatelli Nunes, F., Vákár, M.: **Automatic differentiation for ML-family languages: Correctness via logical relations.** Mathematical Structures in Computer Science. 34, No. 8, 747–806 (60 pages).
- Lucatelli Nunes, F., Vákár, M.: **CHAD for Expressive Total Languages.** Math. Struct. Comput. Sci. 33, No. 4–5, 311–426. 2023. (*116 pages*).
- Lucatelli Nunes, F., Prezado, R., Sousa, L.: **Cauchy Completeness, Lax Epimorphisms, and Effective Descent for Split Fibrations.** Bull. Belg. Math. Soc. - Simon Stevin 30, No. 1, 130–139. 2023. (*10 pages*).
- Lucatelli Nunes, F.: **Semantic factorization and descent.** Appl. Categ. Struct. 30, No. 6, 1393–1433. 2022. (*41 pages*).
- Lucatelli Nunes, F., Sousa, L.: **On lax epimorphisms and the associated factorization.** J. Pure Appl. Algebra 226, No. 12, 27 p. 2022. (*27 pages*).
- Lucatelli Nunes, F.: **Pseudo-Kan Extensions and Descent Theory.** Theory Appl. Categ. 33, 390–444. 2018. (*35 pages*).
- Lucatelli Nunes, F.: **On biadjoint triangles.** Theory Appl. Categ. 31, 217–256. 2016. (*40 pages*).

PREPRINTS

- Lucatelli Nunes, F., Plotkin, G., Vákár, M.: **Unraveling the iterative CHAD.** arXiv:2505.15002. 2025. (*57 pages*).
- Lucatelli Nunes, F., Prezado, R.: **Functors Preserving Effective Descent Morphisms.** arXiv:2410.22876. 2024. (*17 pages*).
- Lucatelli Nunes, F., Vákár, M.: **Free doubly-infinitary distributive categories are cartesian closed.** arXiv:2403.10447. 2024. (*16 pages*).
- Lucatelli Nunes, F.: **Freely generated n-categories, coinserters and presentations of low dimensional categories.** arXiv:1704.04474. 2017. (*56 pages*).

OTHER JOURNAL PUBLICATIONS

- Prezado, R., Lucatelli Nunes, F.: **Generalized multicategories: change-of-base, embedding, and descent**. Appl. Categ. Struct. Volume 32, 35. 2024. (*89 pages*).
- Lucatelli Nunes, F., Prezado, R., Vákár, M.: **Free extensivity via distributivity**. Portugaliae Mathematica, 82, no. 1/2, pp. 177–204. 2025. (*28 pages*).
- Clementino, M.M., Lucatelli Nunes, F., Prezado, R.: **Lax comma categories: cartesian closedness, extensivity, topologicity, and descent**. Theory Appl. Categ. 41, 516–530. 2024. (*15 pages*).
- Clementino, M.M., Lucatelli Nunes, F.: **Lax comma 2-categories and admissible 2-functors**. Theory Appl. Categ. 40, 180–226. 2024. (*47 pages*).
- Clementino, M.M., Lucatelli Nunes, F.: **Lax comma categories of ordered sets**. Quaest. Math. 46, Part Suppl. 1, 145–159. 2023. (*14 pages*).
- Prezado, R., Lucatelli Nunes, F.: **Descent for internal multicategory functors**. Appl. Categ. Struct. 31, 11. 2023. (*18 pages*).
- Lucatelli Nunes, F.: **Descent data and absolute Kan extensions**. Theory Appl. Categ. 37, 530–561. 2021. (*32 pages*).
- Lucatelli Nunes, F.: **Pseudoalgebras and non-canonical isomorphisms**. Appl. Categ. Struct. 27, No. 1, 55–63. 2019. (*9 pages*).
- Lucatelli Nunes, F.: **On lifting of biadjoints and lax algebras**. Categ. Gen. Algebr. Struct. Appl. 9, No. 1, 29–58. 2018. (*30 pages*).
- Lucatelli Nunes, F.: **Espaços não reversíveis**. (Non-reversible spaces). Revista Matemática Universitária, v. 48, n. 49, p. 22–26. 2012. (*5 pages*).

WORKSHOP AND OTHER PUBLICATIONS

- Lucatelli Nunes, F., Plotkin, G., Vákár, M.: **Homomorphic Reverse Differentiation of Iteration**. LAFI 2024 (Tenth Workshop on Languages for Inference at POPL, 2024). 2024. (*2 pages*).
- Lucatelli Nunes, F., Vákár, M.: **Logical Relations for Partial Features and Automatic Differentiation Correctness**. arXiv:2210.08530. 2022. and Oberwolfach Preprints, OWP-2023-09. 2023. (*21 pages*).
- Lucatelli Nunes, F.: **Pseudomonads and Descent**. PhD Thesis, University of Coimbra. 2017. (*209 pages*).

UNDERGRADUATE AND GRADUATE MONOGRAPHS

- Lucatelli Nunes, F.: **Basic Concepts of Topology for Topological Dynamics (in Portuguese)**. 2009. (*116 pages*).
- Lucatelli Nunes, F.: **Basic Topological Dynamics and Basic Applications to Number Theory (in Portuguese)**. 2009. (*57 pages*).

Grants and Fellowships

RESEARCH FELLOWSHIPS (PI)

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| 2022 | Oberwolfach Leibniz Fellowship , Grothendieck Descent Theory, Galois Theory, and Presentations of Categorical Structures <ul style="list-style-type: none"> • PI: Fernando Lucatelli Nunes. • Participants: Maria Manuel Clementino, Rui Prezado, Lurdes Sousa, Matthijs Vákár. |
| 2023 | Fields Research Fellowship , Grothendieck Descent Theory, Fibrations, and Factorizations <ul style="list-style-type: none"> • PI: Fernando Lucatelli Nunes. • Participants: Walter Tholen. |

SELECTED PARTICIPATION IN RESEARCH PROJECTS AND GRANTS

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| 2025 – present | ERC Starting Grant FoRECAST , Formalised Reasoning about Expectations: Composable, Automated, Speedy, Trustworthy <ul style="list-style-type: none"> • PI/Grant holder: Matthijs Vákár. • My role: Member of the formal methods team, contributing to the design of principled algorithms grounded in categorical structures, and involved in the (co-)supervision of PhD students. |
| 2025 – present | DFG Project HOMBRe , Higher-Order Monad-based Programming and Reasoning <ul style="list-style-type: none"> • PI/Grant holder: Sergey Goncharov. • My role: Postdoctoral Fellow, contributing to the core research themes of the project. |
| 2021 – 2024 | NWO Veni Grant (VI.Veni.202.124) , Correct Expressive Differential Programming: Laying the Mathematical Foundations for Tomorrow’s Machine Learning <ul style="list-style-type: none"> • PI/Grant holder: Matthijs Vákár. • My role: Led the work on Iterative CHAD, contributing to the formal methods component of the project. Additionally, established the correctness of dual-numbers-based (forward- and reverse-mode) automatic differentiation for languages with recursive types. |
| 2020 – 2022 | Marie Skłodowska-Curie Grant , Principled AD Program Transformations <ul style="list-style-type: none"> • PI/Grant holder: Matthijs Vákár. • My role: Developed the correct extension of reverse-mode automatic differentiation (CHAD) program transformations to expressive total languages. |
| 2011 – 2011 | Project: Nonlinear Analysis and Applications (CNPq) , Group Project (University of Brasília), funded by CNPq <ul style="list-style-type: none"> • Project title: Nonlinear Analysis and Applications. • My role: Undergraduate student; led the subproject “Topological Dynamics and Applications in Number Theory” and authored two monographs on the topic. |

Selected Graduate Supervision and Mentorship

PHD STUDENTS (THESIS SUPERVISION)

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| 2026 – present | Utrecht University , Department of Information and Computing Sciences <ul style="list-style-type: none">• Student: Diogo Simm.• Research area: Programming language semantics and category theory.• Co-supervision with Dr. Matthijs Vákár and Prof. Gabriele Keller.• Funded under the ERC Starting Grant FoRECAST. |
| 2025 – present | University of Coimbra , Joint University of Coimbra–University of Porto PhD Programme in Mathematics <ul style="list-style-type: none">• Student: Néot Choserot.• Research area: Generalised categorical structures.• Funded by the Centre for Mathematics, University of Coimbra. |
| 2020 – 2024 | University of Coimbra , Joint University of Coimbra–University of Porto PhD Programme in Mathematics <ul style="list-style-type: none">• Student: Rui Rodrigues de Abreu Fernandes Prezado.• Thesis: <i>Some Aspects of Descent Theory and Applications</i>.• Research area: Generalised multicategories, and Grothendieck descent theory.• Co-supervision with Prof. Maria Manuel Clementino.• Funded by FCT, Portugal.• PhD awarded <i>summa cum laude</i> (January 2024). |

MASTER'S STUDENTS (THESIS SUPERVISION)

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| 2025 | IMPA, Rio de Janeiro, Brazil , Master's Programme in Pure Mathematics <ul style="list-style-type: none">• Student: Diogo Simm.• Co-supervised with Prof. Henrique Bursztyn and Dr. Matthijs Vákár.• Research area: Programming language semantics, automatic differentiation, and category theory.• Degree awarded with distinction (grade A). |
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RESEARCH-BASED MASTER'S PROJECT SUPERVISION

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| 2025 | Utrecht University, The Netherlands , Scientific Methods in Computing Science (Programming Languages) <ul style="list-style-type: none">• Supervision of multiple small-group research projects at Master's level.• Topics included symbolic and automatic differentiation, program transformations, and categorical semantics.• Representative projects:<ul style="list-style-type: none">– <i>Calculating the Symbolic Derivative using Automatic Differentiation</i>.– <i>Lambda-Set Specialisation for Haskell</i>.– <i>Dual-Numbers Automatic Differentiation with Inductive Types</i>. |
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Selected Undergraduate Supervision

UNDERGRADUATE THESIS AND CAPSTONE SUPERVISION

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| 2025 – present | Utrecht University , BSc Thesis in Artificial Intelligence <ul style="list-style-type: none">• Student: Sander Santema.• Thesis topic: A local defunctionalisation transformation for the simply typed lambda calculus.• Areas: Categorical semantics, program transformation, defunctionalisation, functional programming, Haskell, and generalised algebraic data types. |
| 2023 | Utrecht University , BSc Capstone Project in Computer Science <ul style="list-style-type: none">• Students: Cérine Toubert, Oscar Mensink, Kasper Peeters, Nigel Strachan, Joost Laan, Jippe Hoogeveen, Gust Verhees, Nasr Antar, and Artur Salek.• Project: Design and implementation of a software system for identifying optimal collaboration matches within organisations, based on interaction data and weighted optimisation criteria.• Client: Dr. Ioanna Lykourantzou• Areas: Optimisation algorithms, software engineering, and web-based systems. |

UNDERGRADUATE SOFTWARE PROJECTS

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| 2021–2022 | Utrecht University , Game Technology <ul style="list-style-type: none">• Supervision of nine undergraduate students organised into three project teams.• Project: Design and development of a game implemented in C#. |
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Teaching and Public Engagement

SELECTED TEACHING EXPERIENCE

2020 – 2025	Utrecht University, The Netherlands, Department of Information and Computing Sciences • Lecturer and Graduate Teaching Assistant (PhD appointment). • Functional Programming — tutorials and assessment. • Logic for Computer Science — tutorials and project supervision. • Graphics for Computer Science (including linear algebra components). • Supervision of BSc Computer Science capstone software projects. • Coaching and supervision of undergraduate Game Technology projects. • Recipient of the Utrecht University Distinguished Teaching Award (2023).
2023 – 2024	Utrecht University, The Netherlands, Department of Information and Computing Sciences • Main Lecturer, BSc Computer Science. • Course: <i>Logic for Computer Science</i>. • Topics included inference systems, natural deduction, semantics, and Hoare logic.
2020	University of Porto, Portugal, Joint PhD Programme in Mathematics (Coimbra–Porto) • Main Lecturer. • Course: <i>Categories in Algebra and Topology</i>. • Topics included enriched categories, monads, and generalised multicategories.
2018 – 2019	University of Coimbra, Portugal, Department of Mathematics • Invited Assistant Professor. • Course: <i>Topology and Linear Analysis</i> (co-taught with Prof. Maria Manuel Clementino).
2017 – 2020	University of Coimbra, Portugal, Department of Mathematics • Instructor, Delfos Project. • Advanced mathematics teaching for secondary-school students preparing for national and international mathematics competitions.
2008 – 2012	University of Brasília, Brazil, Department of Mathematics • Undergraduate Teaching Assistant. • Courses included differential and integral calculus, real analysis, linear algebra, and algebra I.
2006 – 2008	Curso Dínatos, Brasília, Brazil, Secondary Education • Secondary Mathematics Teacher. • Mathematics instruction for students preparing for national university entrance examinations.
2005 – 2006	EPCAR, Brazilian Air Force, Brazil, Secondary Education • Student Teaching Assistant. • Mathematics instruction for pre-cadet students.

OUTREACH AND PUBLIC ENGAGEMENT

2017 – 2020

Delfos Project, Department of Mathematics, University of Coimbra, Portugal

- Contributor to the Delfos Project, a mathematics education programme for secondary-school students preparing for international mathematics competitions.
- Served as Portuguese representative to the International Mathematical Olympiad (IMO) in 2020.

2008 – 2012

Scientific Initiation (CNPq), Brazil

- Undergraduate member of the research group *Teoria de Lie e Dinâmica*.
- Contributed to the outreach blog <https://liedinamica.wordpress.com/>.
- Authored three educational monographs in Portuguese on Lie groups, topology for dynamical systems, and applications of topological dynamics to number theory.
- Published an expository article in *Revista Matemática Universitária* (Portuguese).

2011 – 2012

Brazilian Society for Scientific Progress (SBPC), Brazil

- Presented multiple popular-science posters in mathematics at national SBPC conferences.
- Topics included game theory, homotopy theory, category theory, topology, and dynamical systems.
- All contributions were published in the conference annals (in Portuguese).

Programming Languages and Methodological Expertise

Research and supervisory experience with functional (Haskell, OCaml), multi-paradigm (Python, C#), and web-oriented (JavaScript/TypeScript) programming languages, primarily in the context of research prototypes, program transformations, and teaching.

Evaluation Committees

DOCTORAL EXAMINATION COMMITTEES

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| 2024 | <p>PhD Defense Committee, Joint University of Coimbra–University of Porto
PhD Programme in Mathematics, Portugal</p> <ul style="list-style-type: none">• Candidate: Rui Rodrigues de Abreu Fernandes Prezado.• Thesis: <i>Some Aspects of Descent Theory and Applications</i>.• Role: Supervisor and examiner. |
| 2023 | <p>PhD Defense Committee, Joint University of Coimbra–University of Porto
PhD Programme in Mathematics, Portugal</p> <ul style="list-style-type: none">• Candidate: Carlos Miguel Alves Fitas.• Thesis: <i>On Lax Idempotent Monads in Topology</i>.• Role: Main examiner and opponent. |

DOCTORAL PROJECT EVALUATION

Member of doctoral project evaluation committees within the Joint University of Coimbra–University of Porto PhD Programme in Mathematics (2019), assessing research proposals in category theory and topology, including projects on lax idempotent monads and Grothendieck descent theory. Participated as examiner and supervisor in the formal evaluation of doctoral research plans.

OTHER ACADEMIC EVALUATIONS

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| 2025 | <p>Bachelor's Thesis Proposal Evaluation, BSc in Artificial Intelligence,
Utrecht University, The Netherlands</p> <ul style="list-style-type: none">• Role: Second reader.• Topic: <i>Generative AI as a Study Tool in Higher Education</i>. |
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Academic Service and Governance

RECRUITMENT AND GOVERNANCE

2025	PhD Recruitment Committee Member , ERC Starting Grant FoRECAST • Member of the hiring committee for three PhD positions under the ERC Starting Grant project <i>Formalised Reasoning about Expectations: Composable, Automated, Speedy, Trustworthy</i>.
2023 – 2025	Member, Informatics Research Advisory Committee (IRAC) , Utrecht University, The Netherlands.
2020	Programme Committee Member , Joint University of Coimbra–University of Porto PhD Programme in Mathematics, Portugal.

EDITORIAL AND REFEREEING ACTIVITY

Referee for leading international journals and conferences in mathematics and computer science, including but not limited to: *ACM SIGPLAN Symposium on Principles of Programming Languages (POPL)*, *Advances in Mathematics*, *Applied Categorical Structures*, *Homology, Homotopy and Applications*, *Journal of Pure and Applied Algebra*, *Mathematical Structures in Computer Science*, *Order*, *Quaestiones Mathematicae*, *Theory and Applications of Categories*, and *Topology and its Applications*.

ORGANISATION OF SCIENTIFIC EVENTS AND SEMINARS

2025 – present	Co-organiser, Online Algebra, Logic and Topology Seminar , University of Coimbra, Portugal.
2019 – 2023	Co-organiser, Algebra, Logic and Topology Seminar , University of Coimbra, Portugal.
2022	Organising Committee Member, XIII Portuguese Category Seminar , University of Coimbra, Portugal.
2019	Organising Committee Member, XII Portuguese Category Seminar , University of Coimbra, Portugal.
2018	Organiser, Workshop on Algebra, Logic and Topology , University of Coimbra, Portugal.
2018	Organiser, Workshop: Logic for Children , UNILOG 2018, Vichy, France.

Talks

SELECTED TALKS (INVITED AND/OR REFEREED)

- Leiden University, The Netherlands, 2024
University of Santiago de Compostela, Spain, 2024
University of Coimbra, Portugal, 2024
University of Coimbra, Portugal, 2024
University of Coimbra, Portugal, 2023
Université catholique de Louvain, Belgium, 2023
Lorentz Center, Leiden, Netherlands, 2022
University of Coimbra, Portugal, 2022
University of Primorska, Slovenia, 2021
University of Coimbra, Portugal, 2019
University of Edinburgh, Scotland, 2019
Stellenbosch University, South Africa, 2019
University of Azores, 2018

Masaryk University, Brno, Czech Republic, 2018
Universidade de Santiago de Compostela, Spain, 2018
Université catholique de Louvain, Belgium, 2017
University of Coimbra, Portugal, 2017
University of Coimbra, Portugal, 2016
University of Aveiro, Portugal, 2015
- 109th Peripatetic Seminar on Sheaves and Logic**, The semantic lax descent factorization of a functor.
International Conference in Category Theory 2024, Doubly-infinitary distributive categories
SUMTOPO 2024, Topology and Logic in Computer Science, Homomorphic reverse differentiation for partial features
38th Summer Conference on Topology and its Applications, Free distributive and extensive categories
XIV Portuguese Category Seminar, Freely generated bicartesian categories and exponentials
International Conference in Category Theory 2023, Spanning the tale of “Monades et descente”
New Challenges in Programming Language Semantics, Categorical Approach to Semantic Logical Relations (for partial features and recursive types).
XIII Portuguese Category Seminar, CHAD: elementary categorical aspects of automatic differentiation
8th European Congress of Mathematics, Eilenberg-Moore, Kleisli and descent factorizations.
XII Portuguese Category Seminar, Lax comma 2-categories.

International Conference on Category Theory 2019, Descent and Monadicity.
Categorical Structures in Algebra, Topology and Logic, Eilenberg-Moore, Kleisli and descent factorization.
International Conference on Category Theory (plenary), Aspects of Descent via Bilimits.
103rd Peripatetic Seminar on Sheaves and Logic, Biadjoint Triangles and Applications.
102nd Peripatetic Seminar on Sheaves and Logic, Non-canonical isomorphisms.

5th Workshop on Categorical Methods in Non-Abelian Algebra, Biadjoint Triangles.
XI Portuguese Category Seminar, Freely generated n-categories and coinserter.
4th Workshop on Categorical Methods in Non-Abelian Algebra, Bilimits Commutativity and Descent
International Conference on Category Theory 2019, Kan extensions and descent theory.

(Invited 1-Hour Seminar) TALKS

IMPA, Brazil, 2025	Algebra Seminar , Biadjoint Triangles and Coherence
University of Cape Town, South Africa, 2025	Category Theory Seminar , Bifibrations, Spans, and the Bénabou-Roubaud Theorem
Stellenbosch University, South Africa, 2025	Category Theory Seminar , Two-dimensional monad theory, pseudodistributive laws, and doubly-infinitary distributive and extensive categories as pseudoalgebras
University of Cape Town, South Africa, 2025	Category Theory Seminar , Two-dimensional monad theory, and free extensive completions
University of Coimbra, Portugal, 2024	Algebra, Logic and Topology Seminar , Unraveling the iterative CHAD.
University of Coimbra, Portugal, 2022	Algebra, Logic and Topology Seminar , Descent for Split Fibrations.
Utrecht University, Netherlands, 2021	Software Technology Group Seminar , Automatic differentiation of expressive total functional languages.
University of Coimbra, Portugal, 2019	Seminar of the Category Theory Group at the University of Coimbra , On monadicity and descent.
University of Coimbra, Portugal, 2019	Algebra, Logic and Topology Seminar , Functorial Semantics and Descent II.
University of Coimbra, Portugal, 2018	Algebra, Logic and Topology Seminar , Functorial Semantics and Descent.
University of Coimbra, Portugal, 2018	Algebra, Logic and Topology Seminar , Biadjoint Triangles and Applications.
University of Coimbra, Portugal, 2018	Algebra, Logic and Topology Seminar , Towards semi left exactness and Janelidze-Galois within the lax idempotent context.
University of Coimbra, Portugal, 2016	Algebra, Logic and Topology Seminar , The deficiency of categories.
University of Coimbra, Portugal, 2015	Algebra, Logic and Topology Seminar , Biadjoint triangles, descent and coherence.
Universidade do Porto, Portugal, 2016	Research Seminar Program , Kan construction of adjunctions.
University of Coimbra, Portugal, 2016	Research Seminar Program , Homotopy Excision.
University of Coimbra, Portugal, 2013	Joint PhD Program UC-UP Seminar , Every prespectrum represents a homology theory.
Universidade de Brasília, Brazil, 2010	Math Seminar , Topological Dynamics and Applications in Number Theory (in Portuguese).
Universidade de Brasília, Brazil, 2010	Math Seminar , Dynamical Proof of the Van der Waerden Theorem (in Portuguese).

WORKSHOP TALKS

LAFI, London, UK	LAFI 2024 (Tenth Workshop on Languages for Inference at POPL, 2024) , Homomorphic Reverse Differentiation of Iteration.
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Additional Information

EARLY-CAREER FELLOWSHIPS AND SCHOLARSHIPS

2020 – 2025	PhD Fellowship (Promovendus) , Department of Information and Computing Sciences, Utrecht University, The Netherlands.
2013 – 2017	PhD Fellowship , National Council for Scientific and Technological Development (CNPq), Brazil.
2012 – 2013	Doctoral Research Grant , Centre for Mathematics, University of Coimbra, Portugal.
2009 – 2012	Undergraduate and Master's Scholarships , National Council for Scientific and Technological Development (CNPq) and IMPA, Brazil.
2005	Selective Pre-University Academic Training , EPCAR — Brazilian Air Force Preparatory School for Aeronautics, Brazil.

NATURAL LANGUAGES

Portuguese and English